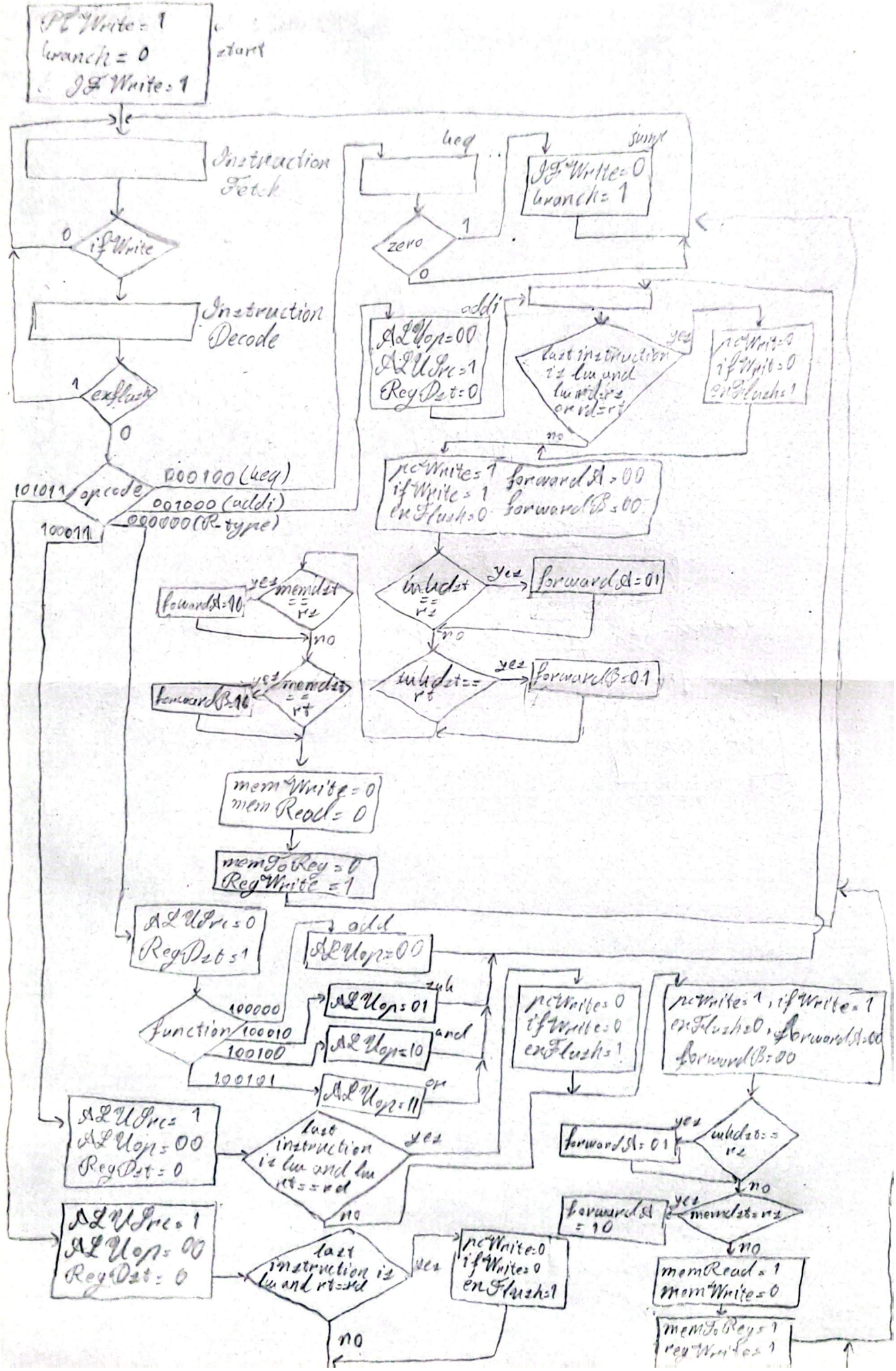
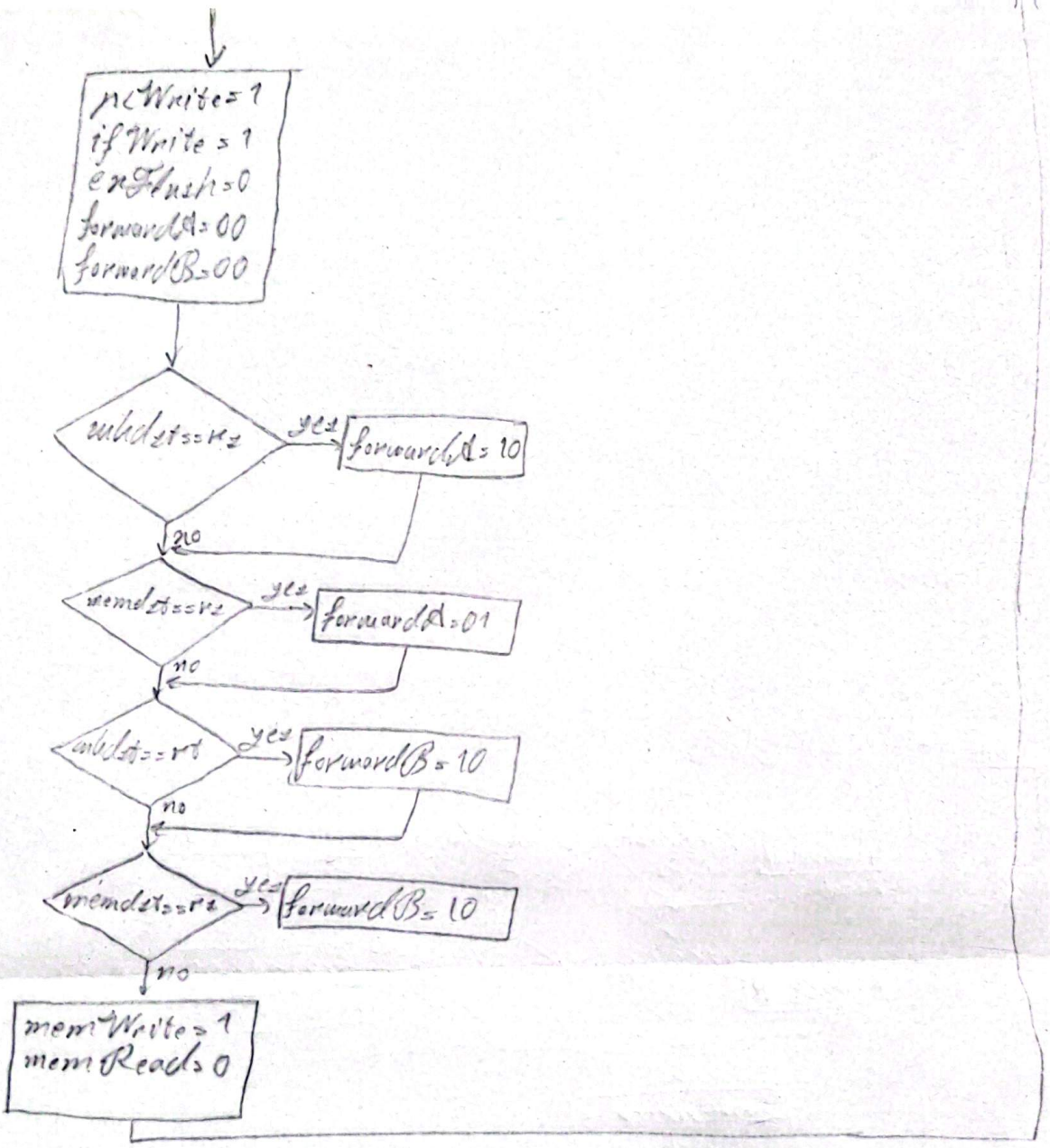

PCWrite = 1
branch = 0
IFWrite = 1
start
Instruction Fetch
ifWrite
Instruction Decode
exflush
opcode
101011
100011
000100 (beq)
001000 (addi)
000000 (R-type)
beq
zero
IFWrite = 0
branch = 1
addi
ALUop = 00
ALUSrc = 1
RegDst = 0
last instruction is lw and lw rt = rs or rd = rt
pcWrite = 0
ifWrite = 0
exFlush = 1
pcWrite = 1
ifWrite = 1
exFlush = 0
forwardA = 00
forwardB = 00
memRd == rs
forwardA = 10
idRd == rs
forwardA = 01
memRd == rt
forwardB = 10
idRd == rt
forwardB = 01
memWrite = 0
memRead = 0
memToReg = 0
RegWrite = 1
ALUSrc = 0
RegDst = 1
add
ALUop = 00
Function
100000
100010
100100
100101
ALUop = 01
sub
ALUop = 10
and
ALUop = 11
or
pcWrite = 0
ifWrite = 0
exFlush = 1
pcWrite = 1, ifWrite = 1
exFlush = 0, forwardA = 00
forwardB = 00
ALUSrc = 1
ALUop = 00
RegDst = 0
last instruction is lw and lw rt == rd
forwardA = 01
idRd == rs
forwardA = 10
memRd == rs
memRead = 1
memWrite = 0
memToReg = 1
regWrite = 1
last instruction is lw and rt = rd
pcWrite = 0
ifWrite = 0
exFlush = 1
yes
no


pcWrite=1
ifWrite=1
exFlush=0
forwardA=00
forwardB=00
wbdst==rs
yes
forwardA=10
no
memdst==rs
yes
forwardA=01
no
wbdst==rt
yes
forwardB=10
no
memdst==rt
yes
forwardB=10
no
memWrite=1
memRead=0